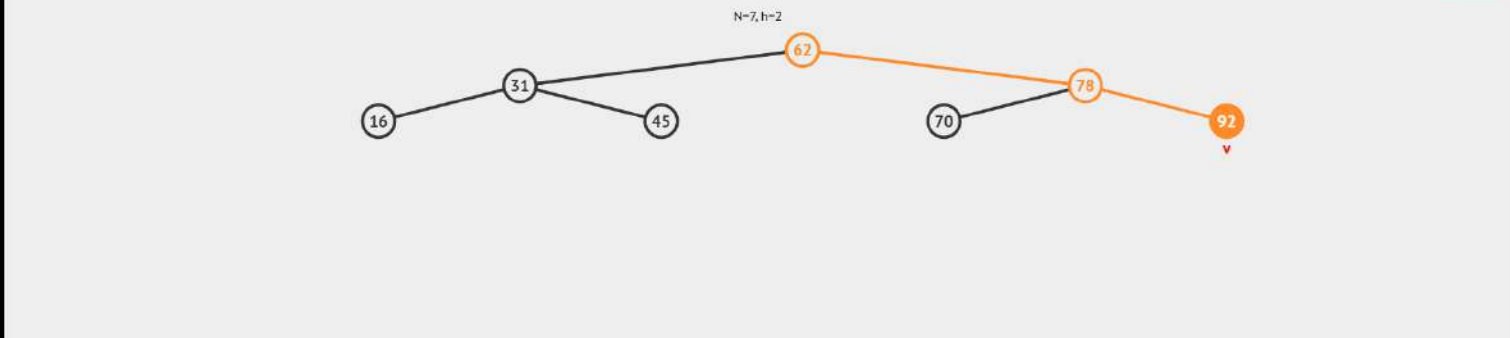




Insert(62, 31, 78, 16, 45, 70, 92)

- Toggle BST Layout
- Create
- Search(v)
- Insert(v)
- Remove(v)
- Predec./Successor(v)
- Select(k)
- Traverse(root)

Inorder(root) Preorder(root) Postorder(root)

```
92 has been inserted!  
  
if insertion point is found  
  create new vertex  
if value to be inserted < this key  
  go left  
else if value to be inserted > this key  
  go right  
else increment frequency
```

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VISUALGO.NET / en / bst BINARY SEARCH TREE AVL TREE

Exploration Mode LOGIN

N=7,h=2

16

31

45

62
root

70

78

92

Inorder Traversal

Inorder traversal of the whole BST is complete.
Visitation sequence: 16,31,45,62,70,78,92.

```
if this is null
  return
Inorder(left)
visit this
Inorder(right)
```

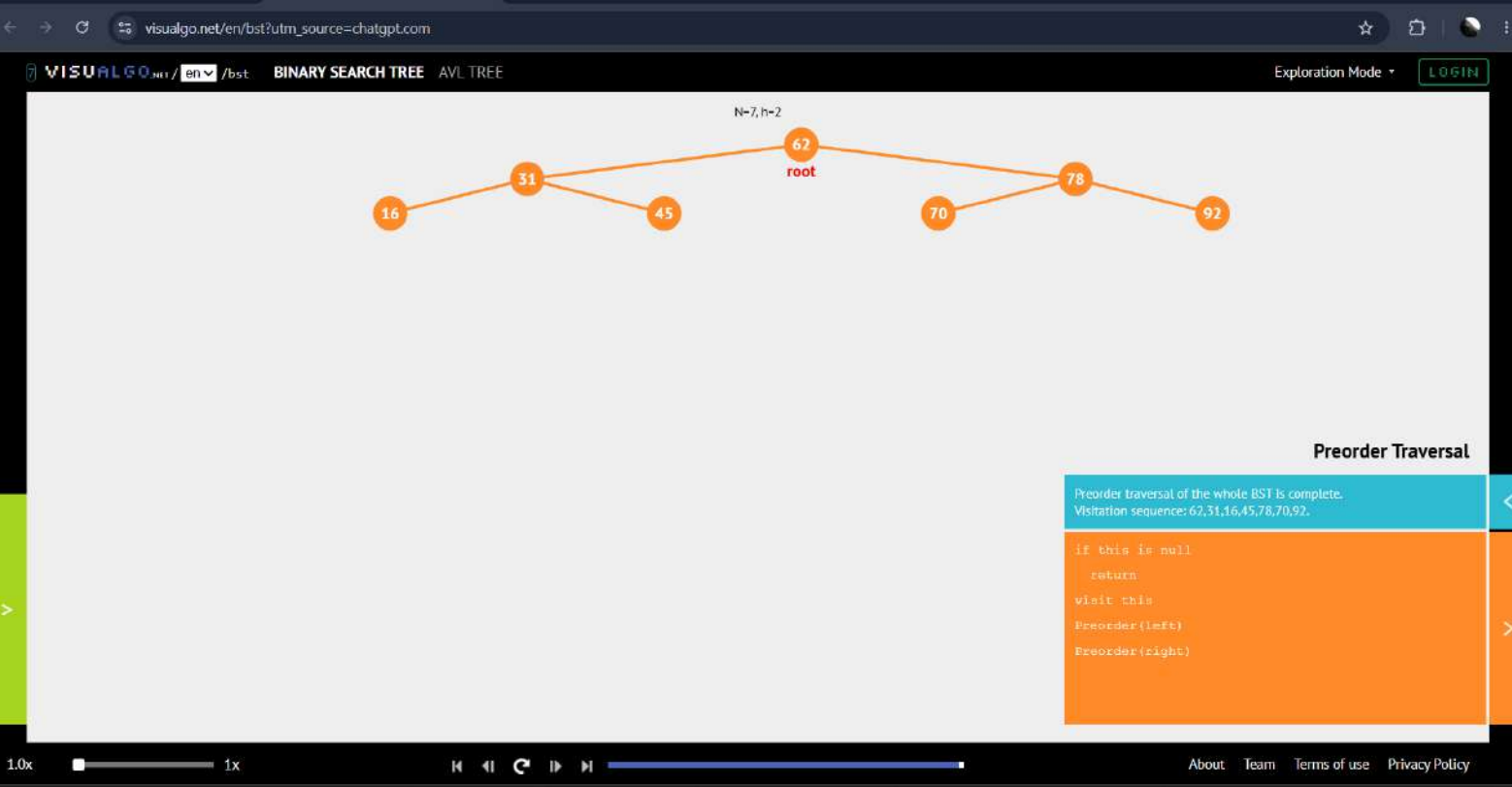
1.0x 1x

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=7, h=2

62

root

16

31

45

70

78

92

Postorder Traversal

Postorder traversal of the whole BST is complete.
Visitation sequence: 16,45,31,70,92,78,62.

```
if this is null
  return
Postorder(left)
Postorder(right)
visit this
```

1.0x

1x

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VISUALGO.NET

bst

BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=3, h=1

20

40

50

Insert(20, 50, 40)

The tree is balanced.

```
insert v
check balance factor of this and its children
case1: this.rotateRight
case2: this.left.rotateLeft, this.rotateRight
case3: this.rotateLeft
case4: this.right.rotateRight, this.rotateLeft
this is balanced
```

<

Toggle BST layout

Create

Search(v)

Insert(v)

Remove(v)

Predec./Successor(v)

Select(k)

Traverse(root)

Inorder(root)

Preorder(root)

Postorder(root)

1.0x

1x

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/bst

BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=3, h=1

20

40
root

50

Preorder Traversal

Preorder traversal of the whole BST is complete.
Visitation sequence: 40,20,50.

```
if this is null
  return
visit this
Preorder(left)
Preorder(right)
```

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VISUALGO.NET / en / bst

BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=7,h=2

17

33

45

59

69

77

92

Insert(59, 33, 77, 17, 45, 69, 92)

92 has been inserted!

if insertion point is found

create new vertex

if value to be inserted < this key

go left

else if value to be inserted > this key

go right

else increment frequency

1.0x

1x

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VISUALGO.NET /en/ /bst BINARY SEARCH TREE AVL TREE Exploration Mode LOGIN

N=7, h=2

```
graph TD; 59((59 root)) --- 33((33)); 59 --- 77((77)); 33 --- 17((17)); 33 --- 45((45)); 77 --- 69((69)); 77 --- 92((92));
```

Inorder Traversal

Inorder traversal of the whole BST is complete.
Visitation sequence: 17,33,45,59,69,77,92.

```
if this is null
  return
inorder(left)
visit this
inorder(right)
```

1.0x 1x ⏮ ⏪ ⏩ ⏭ 🔍

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=7, h=2

17

33

45

59

69

77

Insert(59, 33, 77, 17, 45, 69, 92)

92 has been inserted!

insert v

check balance factor of this and its children

case1: this.rotateRight

case2: this.left.rotateLeft, this.rotateRight

case3: this.rotateLeft

case4: this.right.rotateRight, this.rotateLeft

this is balanced

1.0x

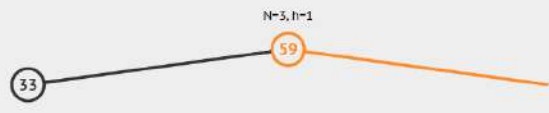
1x

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Insert(59, 33, 77, 17, 45, 69, 92)

```
77 has been inserted!  
if insertion point is found  
  create new vertex  
if value to be inserted < this key  
  go left  
else if value to be inserted > this key  
  go right  
else increment frequency
```



92 has been inserted!

```
if insertion point is found
```

```
create new vertex
```

```
if value to be inserted < this key
```

```
else if value to be inserted > this key
```

also increment frequency

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=7,h=2

17

33

45

59

69

77

92

Inorder Traversal

The right child of vertex with value 45 is empty.
Return empty.

if this is null
return

Inorder(left)
visit this
Inorder(right)

1.0x

1x

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ENG
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10:43
13-08-2026

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=7, h=2

59
root

17

33

45

69

77

92

Inorder Traversal

Inorder traversal of the whole BST is complete.
Visitation sequence: 17,33,45,59,69,77,92.

```
if this is null
  return
Inorder(left)
visit this
Inorder(right)
```

1.0x

1x

1043

13-08-2026

Search

1043

13-08-2026

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BINARY SEARCH TREE

AVL TREE

Exploration Mode

LOGIN

N=7, h=2

```
graph TD; 59((59)) --- 33((33)); 59 --- 77((77)); 33 --- 17((17)); 33 --- 45((45)); 77 --- 69((69)); 77 --- 92((92)); style 17 fill:#ff9933; style 33 fill:#ff9933; style 59 fill:#ff9933;
```

Preorder Traversal

The left child of vertex with value 17 is empty.
Return empty.

if this is null
return

visit this
Preorder(left)
Preorder(right)

1.0x

1x

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VISUALGO.NET /en/ BINARY SEARCH TREE AVL TREE Exploration Mode LOGIN

N=7, h=2

```
graph TD; 59((59 root)) --- 33((33)); 59 --- 77((77)); 33 --- 17((17)); 33 --- 45((45)); 77 --- 69((69)); 77 --- 92((92));
```

Preorder Traversal

Preorder traversal of the whole BST is complete.
Visitation sequence: 59,33,17,45,77,69,92.

```
if this is null
  return
visit this
Preorder(left)
Preorder(right)
```

1.0x 1x

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Postorder Traversal

The right child of vertex with value 17 is empty.
Return empty.

```
if this is null  
    return
```

```
Postorder(left)  
Postorder(right)  
visit this
```

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VISUALGO.NET / en / bst BINARY SEARCH TREE AVL TREE Exploration Mode LOGIN

N=7, h=2

```
graph TD; 59((59 root)) --- 33((33)); 59 --- 77((77)); 33 --- 17((17)); 33 --- 45((45)); 77 --- 69((69)); 77 --- 92((92));
```

Postorder Traversal

Postorder traversal of the whole BST is complete.
Visitation sequence: 17,45,33,69,92,77,59.

```
if this is null:  
    return  
Postorder(left)  
Postorder(right)  
visit this
```

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Search

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Tree Simulator Recommendation x Binary Search Tree, AVL Tree - x Binary Search Tree, AVL Tree - x visualising data structures and x

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Ask Gemini

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Exploration Mode LOGIN

N=3, h=2

```
graph LR; 25((25)) --- 50((50)); 50 --- 75((75));
```

bf = -2

bf = -1

Insert(25, 50, 75)

And balance factor of 50 is -1.

```
insert v
check balance factor of this and its children
case1: this.rotateRight
case2: this.left.rotateLeft, this.rotateRight
case3: this.rotateLeft
case4: this.right.rotateRight, this.rotateLeft
this is balanced
```

1.0x 1x

10:44 13-08-2026

